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[22ECCT501]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

SPREAD SPECTRUM TECHNOLOGY

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	What are shift register sequences?	L1	CO1	**02**
	b.	Describe briefly partial-band interference.	L2	CO2	**02**
	c.	What is matched filter acquisition?	L2	CO3	**02**
	d.	Why do we require blind MMSE detector?	L3	CO4	**02**
	e.	Illustrate GPS ranging signals.	L2	CO6	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Explain characteristic polynomials autocorrelations and power spectra.	L2	CO1	**05**
		OR			
	b.	Differentiate general tone interference and gaussian interference.	L3	CO1	**05**
3.	**a.**	Explain partial band interference.	L2	CO2	**05**
		OR			
	b.	Illustrate CPM and CPFSK.	L2	CO2	**05**
4.	**a.**	Explain frequency and timing synchronizations.	L2	CO3	**05**
		OR			
	b.	Discuss rapid timing and serial-search acquisitions.	L2	CO3	**05**
5.	**a.**	Examine blind MMSE detector.	L4	CO4	**05**
		OR			
	b.	Explain minimum mean-square error detection.	L2	CO4	**05**
6.	**a.**	Interpret optimal multiuser rule for synchronous CDMA.	L2	CO5	**05**
		OR			
	b.	Explain multicarrier DS CDMA.	L2	CO5	**05**
7.	**a.**	Illustrate GPS ranging signals.	L2	CO6	**05**
		OR			
	b.	Illustrate GLONASS and GNSS applications.	L2	CO6	**05**

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		PART-C			6 X 10=60
8.	a.	Examine pulsed interference and decoding metrics.	L4	CO1	10
		OR			
	b.	Examine noncoherent systems dispreading with bandpass matched filters.	L4	CO1	10
9.	a.	Illustrate power spectral densities of FH-CPM.	L2	CO2	10
		OR			
	b.	Illustrate digital demodulation of FH-CPFSK.	L2	CO2	10
10.	a.	Interpret the performance of serial-search acquisition and sequential acquisition.	L3	CO3	10
		OR			
	b.	Interpret the consecutive-count double-dwell system.	L3	CO3	10
11.	a.	Examine asynchronous multiuser detectors.	L4	CO4	10
		OR			
	b.	Examine conventional MC transmission and OFDM.	L4	CO4	10
12.	a.	Illustrate multicarrier CDMA applications.	L2	CO5	10
		OR			
	b.	Illustrate transmit diversity and space-time coding in CDMA systems.	L2	CO5	10
13.	a.	Explain general system principles and architecture of GPS.	L2	CO6	10
		OR			
	b.	Explain forward link modulation and reverse link modulation.	L2	CO6	10